

## Abbreviated "order items" list for the V93000/Multilane HSIO system (Smart Scale)

### Order items for the V93000/Multilane HSIO system (per system unless otherwise specified)

| Item                                             | Part number       | Description                                                                                                 | Quantity possible | Comments                                                       |
|--------------------------------------------------|-------------------|-------------------------------------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------|
| <b>Advantest</b>                                 |                   |                                                                                                             |                   |                                                                |
| <b>V93000 Infrastructure and HSIO components</b> |                   |                                                                                                             |                   |                                                                |
| A1                                               | E8028THU          | Test Head upgrade kit for use with family board extension                                                   | 1                 | Must for all installations                                     |
| A2                                               | E8028PSD          | Standard pogo segment for family board extension                                                            | 1 to 16           | Must for all installations                                     |
| A3                                               | E8028HDS          | Hard docking spacer for the family board extension E8028BF                                                  | 1                 | Must for all installations                                     |
| A4                                               | 1400-2000-S       | Pogo Pin replacements (quantity 30) ordered through Tier 1 (support-americas@advantest.com )                | 1                 | For the repair of the pogo segments                            |
| A5                                               | E7016DP           | Coarse alignment pins mounted on the handler docking plate (1 per test cell)                                | 1                 | Only needed if handler hasn't docked to 93K before             |
| A6                                               | E7016ELP          | Extra Long Pins used for docking to a prober only on the docking plate (1 per test cell)                    | 1                 | Only needed if prober hasn't docked to 93K before              |
| A7                                               | E6979LX           | CTH Motorized kit (option - See check page for more info)                                                   | 1                 | Optional                                                       |
| <b>Probe</b>                                     |                   |                                                                                                             |                   |                                                                |
| A11                                              | E8016BBWS         | Direct Dock beam for WSRF/PMUX                                                                              | 1                 | Must for probe                                                 |
| A12                                              | E8028-61270       | RF Kit for direct probe stiffener                                                                           | 1                 | Optional                                                       |
| <b>Multilane</b>                                 |                   |                                                                                                             |                   |                                                                |
| M1                                               | AT93000-TFRAME    | Base Frame for DUT board family board extension (HSIO Card cage)                                            | 1                 | <a href="#">ATE   MultiLane (multilaneinc.com)</a>             |
| M2                                               | AT93000-BASE002   | Base System including ML backplanes, Utility Box, 1900002 Family Board                                      | 1                 | Many items bundled into 1 part number                          |
| M3                                               | AT93000-1900002   | Part of AT93000-BASE002. Family Board                                                                       | 0                 | <a href="#">AT93000-1900002   MultiLane (multilaneinc.com)</a> |
| M4                                               | AT93000-64120     | Part of AT93000-BASE002. Family Board Stiffener                                                             | 0                 | <a href="#">AT93000-1900002   MultiLane (multilaneinc.com)</a> |
| M5                                               | AT93000-TWINN     | Part of AT93000-BASE002. Twinning Frame backplane assembly and cable harnesses                              | 0                 | No website landing page descriptions                           |
| M6                                               | AT93000-UBOX      | Part of AT93000-BASE002. Utility Box Hub - Air/Power/Ethernet                                               | 0                 | <a href="#">AT93000-UBOX   MultiLane (multilaneinc.com)</a>    |
| M7                                               | ML instruments    | Multilane BERTs, DSOs, AWG instruments. See "Multilane Instruments for V93000" on our website               | 1 to 8            | <a href="#">ATE   MultiLane (multilaneinc.com)</a>             |
| M8                                               | AT93000-DIAG2P16C | Diagnostic Kit with 2 POGO and 16 COAX cables. (need 1 at each location that has 1 or more twinning frames) | 1                 | <a href="#">Diagnostic Kit   MultiLane (multilaneinc.com)</a>  |
| M9                                               | AT93000-SMPM-TOOL | Tool to remove pins from the blind mate connector on the DUT bd/probe card (AT93000-POGO)                   | 1 per site        | <a href="#">ATE   MultiLane (multilaneinc.com)</a>             |
| <b>Customer Loadboard</b>                        |                   |                                                                                                             |                   |                                                                |
| C1                                               | AT93000-64150     | Loadboard stiffener assembly for package test                                                               | 1                 | <a href="#">AT93000-64150   MultiLane (multilaneinc.com)</a>   |
| C2                                               | AT93000-64170     | Loadboard stiffener assembly for wafer test                                                                 | 1                 | <a href="#">AT93000-64170   MultiLane (multilaneinc.com)</a>   |
| C3                                               | AT93000-POGO      | SMPM pogo/connector bracket. See AT93000-64150 or AT93000-64170 assembly guide for more info                | 4                 | <a href="#">AT93000-64150   MultiLane (multilaneinc.com)</a>   |
| C4                                               | n/a               | Customer Loadboard                                                                                          | n/a               | Device specific loadboard                                      |
| C5                                               | n/a               | High-speed components, RF switches, cables, and other long lead parts                                       | n/a               | Placeholder reminder for possibly long lead parts              |

## Abbreviated "order items" list for the V93000/Multilane HSIO system (EXA Scale)

### Order items for the V93000/Multilane HSIO system (per system unless otherwise specified)

| Item | Part number       | Description                                                                                                 | Quantity possible | Comments                                                                       |
|------|-------------------|-------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------|
|      |                   | <b>Advantest</b>                                                                                            |                   |                                                                                |
|      |                   | <b>V93000 Infrastructure and HSIO components</b>                                                            |                   |                                                                                |
| A1   | E8028THUX         | Test Head upgrade kit for use with family board extension                                                   | 1                 | Must for all installations                                                     |
| A2   | E8028PSD          | Standard pogo segment for family board extension                                                            | 1 to 16           | Must for all installations                                                     |
| A3   | E8028HDS          | Hard docking spacer for the family board extension E8028BF                                                  | 1                 | Must for all installations                                                     |
| A4   | 1400-2000-S       | Pogo Pin replacements (quantity 30) ordered through Tier 1 (support-americas@advantest.com )                | 1                 | For the repair of the pogo segments                                            |
| A5   | E7016DP           | Coarse alignment pins mounted on the handler docking plate (1 per test cell)                                | 1                 | Only needed if handler hasn't docked to 93K before                             |
| A6   | E7016ELP          | Extra Long Pins used for docking to a prober only on the docking plate (1 per test cell)                    | 1                 | Only needed if prober hasn't docked to 93K before                              |
| A7   | E6979LX           | CTH Motorized kit (option - See check page for more info)                                                   | 1                 | Optional                                                                       |
|      |                   | <b>Probe</b>                                                                                                |                   |                                                                                |
| A11  | E8016BBWS         | Direct Dock beam for WSRF/PMUX                                                                              | 1                 | Must for probe                                                                 |
| A12  | E8028-61270       | RF Kit for direct probe stiffener                                                                           | 1                 | Optional                                                                       |
|      |                   | <b>Multilane</b>                                                                                            |                   |                                                                                |
| M1   | AT93000-TFRAME    | Base Frame for DUT board family board extension (HSIO Card cage)                                            | 1                 | <a href="#">ATE   MultiLane (multilaneinc.com)</a>                             |
| M2   | AT93000-BASE003   | Base System including ML backplanes, Utility Box, xxyzz Family Board                                        | 1                 | Many items bundled into 1 part number                                          |
| M3   | AT93000-3006060   | Part of AT93000-BASE003. Exascale Striped Family Board, RDA job: ADVANTEST-3006060-Rev0                     | 0                 | <a href="#">Exascale Striped Family Board, RDA job: ADVANTEST-3006060-Rev0</a> |
| M4   | AT93000-64120     | Part of AT93000-BASE003. Family Board Stiffener                                                             | 0                 |                                                                                |
| M5   | AT93000-TWINN     | Part of AT93000-BASE003. Twinning Frame backplane assembly and cable harnesses                              | 0                 |                                                                                |
| M6   | AT93000-UBOX      | Part of AT93000-BASE003. Utility Box Hub - Air/Power/Ethernet                                               | 0                 | <a href="#">AT93000-UBOX   MultiLane (multilaneinc.com)</a>                    |
| M7   | ML instruments    | Multilane BERTs, DSOs, AWG instruments. See "Multilane Instruments for V93000" on our website               | 1 to 8            | <a href="#">ATE   MultiLane (multilaneinc.com)</a>                             |
| M8   | AT93000-DIAG2P16C | Diagnostic Kit with 2 POGO and 16 COAX cables. (need 1 at each location that has 1 or more twinning frames) | 1                 | <a href="#">Diagnostic Kit   MultiLane (multilaneinc.com)</a>                  |
| M9   | AT93000-SMPM-TOOL | Tool to remove pins from the blind mate connector on the DUT bd/probe card (AT93000-POGO)                   | 1 per site        | <a href="#">ATE   MultiLane (multilaneinc.com)</a>                             |
|      |                   | <b>Customer Loadboard</b>                                                                                   |                   |                                                                                |
| C1   | AT93000-64150     | Loadboard stiffener assembly for package test                                                               | 1                 | <a href="#">AT93000-64150   MultiLane (multilaneinc.com)</a>                   |
| C2   | AT93000-64170     | Loadboard stiffener assembly for wafer test                                                                 | 1                 | <a href="#">AT93000-64170   MultiLane (multilaneinc.com)</a>                   |
| C3   | AT93000-POGO      | SMPM pogo/connector bracket. See AT93000-64150 or AT93000-64170 assembly guide for more info                | 4                 | <a href="#">AT93000-64150   MultiLane (multilaneinc.com)</a>                   |
| C4   | n/a               | Customer Loadboard                                                                                          | n/a               | Device specific loadboard                                                      |
| C5   | n/a               | High-speed components, RF switches, cables, and other long lead parts                                       | n/a               | Placeholder reminder for possibly long lead parts                              |

| Item | Part Number         | Description                                                                                                                                                                                                                                                                                          |
|------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | N/A                 | Check that all of the "Order" tab checklist Advantest items have been ordered in correct quantities. Choose correct tab Smart Scale or EXA.                                                                                                                                                          |
| 2    | N/A                 | Check that all of the "Order" tab checklist Multilane items have been ordered in correct quantities. Choose correct tab Smart Scale or EXA.                                                                                                                                                          |
| 3    | N/A                 | Check that all of the "Order" tab checklist 3rd party items have been ordered in correct quantities. Choose correct tab Smart Scale or EXA. Must have a Smart Scale Enhanced DUT Scale interface (with hard docking if it is to be docked to a handler or prober) (PN E8029-60122 in EDC Codes file) |
| 4    | E8029D or E8029CD   | E8029D is for STH, E8029CD is for CTH OR                                                                                                                                                                                                                                                             |
| 5    | E8029DD or E8029CDD | Must have a Smart Scale Enhanced DUT Scale interface (no hard docking ) (PN E8029-60121 in EDC Codes file) E8029DD is for STH, E8029CDD is for CTH                                                                                                                                                   |
| 6    | N/A                 | Must be either CTH or STH test head. 6/6/2023 CX and SX is supported on EXA Scale.                                                                                                                                                                                                                   |
| 7    | E6979ABU            | Must have STH test-head extended weight manipulator if STH is used.                                                                                                                                                                                                                                  |
| 8    | E6979LU             | Must have CTH test-head extended weight manipulator if CTH is used. Default on new systems. (See Diagrams tab, diagram 10 for location of PN)                                                                                                                                                        |

|    |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | N/A             | Must have 2 PS1600 channels available for HSIO HW brought up through pogo segment 1A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 10 | N/A             | If probe solution pogo segments 3 and 8 have some reserved pins. (keep out zones on the bottom of the family board)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 11 | N/A             | Check the vertical space stackup to make sure the HSIO card cage & test head will dock to the handler. It may be necessary to raise the handler spacing (replacement, "longer" feet) or lower the V93000 test head with the ESMO kit Adapter kit (z-150 mm)<br>When you install the HSIO card cage it is an immediate 50 lb or more delta. Any combination of weights will not be able to handle this weight delta<br>moving the test head down when the frame is added if the lock on the manipulator is not enabled. Recommendation is to use the lock whenever mounting/dismounting the HSIO card cage. The motor option has a belt that adds enough tension/friction to the setup so that it can handle this weight delta without the test head moving even when unlocked.<br>1. Confirm the standard family board will be sufficient for resources needed.<br>2. Note: 6/6/2023 Exascale Striped Family Board, RDA job: ADVANTEST-3006060-Rev0 in development. |
| 12 | E6979LX         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 13 | AT93000-1900002 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 14 | N/A             | Must use Redhat RHEL7.4 revision is used (or a dual boot with this revision)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 15 | N/A             | Must use Smartest Revision 7.5 or later. For use with the new Communications Link II hardware, this needs to be Smartest 7.5.5 or later.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 16 | N/A             | 6/6/2023 Smartest 8 support under development, check Teamforge project for latest status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 16 | N/A             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 16 | N/A             | Must have 1 physical Ethernet port available to the HSIO test cell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 17 | N/A             | Must have 2 air line drops to the HSIO test cell (1 for V93000, 1 for HSIO card cage)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 18 | N/A             | Must have a site review done for new system installations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 19 | N/A             | Must have the handler/prober docking plates setup to dock to the V93000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 20 | E8016BBWS       | Must have the prober manufacturer to install the direct dock bridge beam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 21 | Model file      | The model file describes the system configuration and is located here: By default, the model file " <b>tester.model</b> " is located in the directory: /etc/opt/hp93000/soc/<br><br>Every time SmarTest starts it creates the file "edc_codes_pon" that holds up-to-date edc code data.<br>The file is located in /var/opt/hp93000/soc_common/diagnosti and can be used to automatically collect information about the current hardware configuration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 22 | EDC_Pon file    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |